

Prediction and Explanation Stability in Class-Incremental Network Traffic Classification

OFRA prediction and method-conditioned explanation auditing

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Abstract

Class-incremental traffic classifiers must learn new classes without an oracle task identifier while retaining earlier knowledge. We evaluate OFRA, a composition of a frozen tabular encoder, low-rank class heads, bounded replay and centroid routing, together with a post-hoc explanation audit. Five-seed prediction results on five independently processed datasets show heterogeneous trade-offs. In a same-data system comparison, Malaya accuracy is 54.68% for OFRA and 40.14% for Balanced Replay50, with signed forgetting of 3.55 and 43.80 percentage points. On ReplayIDS D2, accuracy and Macro-F1 improve while attack recall falls from 99.78% to 85.73%. These comparisons do not isolate architecture from loss, exposure or total memory. A separate five-seed Malaya audit uses a CPU class-batch reconstruction with exact forward values and surrogate gradients. Expected Gradients, Feature Ablation and Gradient x Input yield mean silent-drift rates of 73.98%, 3.60% and 14.34%; three-method ledger-state agreement is 54.0%. Recovered threshold sweeps confirm strong policy dependence. Of 72 checkpoint-class records with recall below 20%, 27 have a historical certified state under at least one method, showing that the ledger does not certify predictive reliability. We therefore treat ETG as an audit record, not an operational governance contribution. The study identifies retention, detection and explanation trade-offs; causal algorithmic advantage, attribution faithfulness and deployment benefit remain unverified.

Keywords: class-incremental learning; network traffic classification; bounded replay; explanation stability; attribution sensitivity; reproducibility.

1. Introduction

1.1 Background and motivation

Network traffic changes as applications, services, attacker behaviour, and collection environments evolve. A fixed classifier therefore becomes incomplete, but repeatedly retraining one shared model can overwrite knowledge of earlier classes. Class-incremental learning addresses this setting by presenting new classes sequentially and evaluating all classes observed so far. For a security-facing system, predictive retention is only one requirement. Analysts may also rely on explanations to understand why a class score was produced, so a stable accuracy value does not by itself guarantee stable decision evidence.

OFRA produces class scores and decisions. The explanation analysis uses stored checkpoints and fixed true-class probes, while ETG records performance and explanation-rule states after inference. This audit does not suppress prediction alerts or control subsequent learning. Its historical certification labels do not imply predictive correctness.

1.2 Problem statement

An incremental classifier can retain aggregate accuracy while minority recall or its attributed feature ranking changes. A useful evaluation must distinguish these outcomes and specify which numerical score is being explained. Source binding alone is insufficient: forward reconstruction, gradient faithfulness and operational utility are separate questions. This study tests predictive retention and explanation stability and examines how attribution choices propagate into a rule-based audit. It does not claim that the current audit governs a deployed detector.

1.3 Research questions and objectives

RQ1: How does oracle-free class-incremental learning affect predictive performance, class-balanced detection, and forgetting across heterogeneous network-traffic datasets?

RQ2: How sensitive are feature-attribution stability measurements to class-incremental updates, attribution method and the permitted class-recall change?

RQ3: How do attribution choices and class performance affect the states of a non-suppressing explanation audit?

RO1 is to evaluate accuracy, Macro-F1, balanced accuracy, attack recall, false-positive rate and forgetting under separate dataset protocols. RO2 is to quantify attribution stability and threshold sensitivity on fixed probes across checkpoints. RO3 is to reproduce method-conditioned audit states and test their relationship with class performance, without equating an explanation-rule pass with safety.

1.4 Contributions and attribution of methods

The empirical contribution is a reproducible account of how retention, minority-class coverage and explanation-rule outcomes differ within a routed class-incremental traffic system. Same-state decision arms expose scoring trade-offs, and three-method, five-seed audits quantify attribution dependence. These measurements identify limitations of the design; they do not establish a new learning algorithm or a beneficial governance controller.

The constituent learning and explanation methods are not new algorithms introduced here. Low-rank adaptation, FT-Transformer, TabM, DP-Means, SHAP, Feature Ablation, and Gradient x Input are existing methods. Family-head integration, the registered score combination, fixed-probe audit protocol, and ledger rules are study-specific design choices. A combination of established components does not by itself demonstrate algorithmic novelty or a causal architectural advantage. Those stronger claims require controlled component comparisons and operational validation. FT-Transformer follows Gorishniy et al. (2021); DP-Means originates from Kulis and Jordan (2012).

2. Related Work

2.1 Continual learning and parameter-efficient adaptation

Catastrophic forgetting motivates regularisation, rehearsal, distillation, exemplar, and parameter-isolation methods in continual learning (French, 1999; Kirkpatrick et al., 2017; Rebuffi et al., 2017). LoRA freezes a base transformation and represents a trainable update with low-rank matrices (Hu et al., 2022). For tabular representation learning, TabM uses parameter-efficient ensembling to produce multiple predictions within one model (Gorishniy et al., 2025), while CatBoost is a gradient-boosted tree method designed to reduce prediction shift in boosting (Prokhorenkova et al., 2018). The present family-specific low-rank head and TabM mean-embedding adapter are project integration choices; neither is presented as a new derivation of the cited methods.

TRIL3 generates pseudorehearsal examples from tabular prototypes rather than retaining old samples (Garcia-Santaclara et al., 2024). PEARL adapts LoRA rank during continual learning (Bhat et al., 2026), while C-LoRA routes low-rank updates through a learnable matrix (Zhang et al., 2025). Their mechanisms are relevant comparators, but the latter studies are not direct reproductions of this flow-feature protocol. Cumulative CatBoost and TabM remain headroom diagnostics rather than continual-learning baselines.

2.2 Explanation drift and IDS governance

CLEX already studies drifting explanations in class-incremental learning (Cossu et al., 2024); the observation that accuracy and explanations can behave differently is therefore not new here. E2D2 monitors explanation-based drift in unsupervised NIDS (Nugraha and Bauschert, 2026), and Maseno et al. (2026) study adversarial-evasion signals. Goncalves and Alomari (2026) evaluate analyst-facing SHAP/LIME, while Adabara et al. (2025) combine explanation and uncertainty. Drift2Act already links drift signals to budgeted interventions and risk assessment (Lamaakal et al., 2026). Our offline ledger provides neither an equivalent intervention study nor a demonstrated advance over these controllers.

Table 1. Positioning against the four recent studies identified during review.

Study	Drift or trust trigger	IDS setting	Gap relative to this work
E2D2 (2026)	Unsupervised explanation-based drift	Unsupervised NIDS	Not legitimate class-incremental model updates; no ETG action ledger
Maseno et al. (2026)	Adversarial SHAP instability	Recurrent multiclass IDS	Adversarial trigger, not benign learning increments; no non-suppressing governance
Goncalves and Alomari (2026)	Human interpretation of SHAP/LIME	AWS GuardDuty	User study rather than longitudinal routed-score stability
Adabara et al. (2025)	Calibration, uncertainty, and SHAP	CICIDS2017	No class-incremental sequence or explanation-state transition ledger

2.3 Research gap

The present scope is a bounded empirical audit of a specific routed traffic classifier. Novelty cannot follow solely from adding LoRA, explanation drift or a ledger to known components. A stronger future contribution would need a falsifiable mechanism and protocol-matched evidence that it improves prediction or resource-constrained repair. Section 5.5 identifies that unresolved test rather than presenting it as a result.

2.4 Routing and unknown-class detection

GCR uses shared frozen geometry to route industrial-image anomaly detectors (Chae et al., 2026). Xie et al. (2024) infer task identity using task-specific batch normalisation and unknown-class outputs in image CIL. VAEMax combines payload features, OpenMax and a VAE rejection stage for intrusion detection (Qiu et al., 2024). These studies motivate explicit routing and open-set controls; their different features and protocols preclude direct numerical comparison with OFRA. No corresponding matched external-baseline result is claimed here.

3. Methodology

3.1 Scope and dataset separation

Each dataset is preprocessed, trained, and evaluated independently; records from different datasets are never pooled. The strict five-seed table uses seeds 1, 2, 3, 4, and 42. NSL-KDD, UNSW-NB15, CIC-IDS-2017, and CSE-CIC-IDS2018 use FT-Transformer 256x4; MalayaNetwork_GT uses FT-Transformer 512x12. These are dataset-specific audits, not a capacity-matched pooled comparison. The CSE-CIC-IDS2018 8+10-epoch campaign completed as DICC Job 402073 and passed the protected SHA-256 manifest check.

Dataset-specific preprocessing removes identifiers, normalises labels, converts features to finite numerical arrays, fixes feature and class order, and constructs class-by-split shards. NSL-KDD retains its official train/test files; its protocol, service, and flag vocabularies are fitted on Task-0 training rows only and frozen, producing 116 features. UNSW-NB15 likewise retains its official 175,341-row training and 82,332-row testing files; its protocol, service, and state vocabularies are fitted on Task-0 training rows only and frozen, producing 186 features. Later unseen categorical values use the registered all-zero block for that field. Malaya uses 31 derived-flow CSVs at frozen revision 384a5927..., removes IP addresses, ports, and timestamps, and holds out one capture per class, producing 77 numerical features. The capture-level Malaya split avoids leakage from correlated rows within one capture.

The current formal prediction protocols use Task-0-only numerical statistics and categorical vocabularies. Numerical standardisation reads only Task-0 training shards, freezes before pretraining, and rejects later updates. NSL-KDD and UNSW-NB15 vocabularies likewise use Task-0 rows only; later unseen values map to all-zero categorical blocks. The strict widths are 116 and 186 features, and the affected five-seed experiments were rerun. Earlier 122/194-feature builders exposed six and eight future-only one-hot columns, affecting about 0.9% of later-task rows. That schema exposure is historical rather than a property of the current prediction table. CIC-IDS-2017, CSE-CIC-IDS2018, and Malaya have no learned categorical vocabulary. This fitting contract does not imply that every separate dataset-construction policy is prospective; the ReplayIDS D2 exception is specified below.

3.2 OFRA architecture and registered decision arms

For Malaya, the shared FT-Transformer maps each numerical row to a 512-dimensional representation using 12 layers, 16 attention heads, head dimension 32, and dropout 0.1. The other formal datasets use the separately registered FT256x4 configuration. After Task-0 training the shared encoder is frozen. Task 0 uses eight epochs and later tasks ten epochs. Each seen class has a rank-8, alpha-16 positive-versus-negative low-rank head. Replay retains at most 50 exemplars per class from a candidate pool of at most 5,000.

The DP-Means router stores up to 32 centroids per class. The capped router fits centroids from at most 3,000 selected training samples per class; the cap limits router fitting cost and majority-class influence, not test-set size. The registered arms are head-only $p(c,x)$, router-only $z(c,x)$, and joint $p(c,x)+0.5z(c,x)$, each with capped and full router variants. Prediction is the global argmax across all seen classes. The joint weight 0.5, cap 3,000, distance quantile 0.3, and centroid limit 32 are study-defined registered settings rather than universal constants.

Each rank- r output-space family head has $2dr+2d+2$ parameters, giving 9,218 parameters at $d=512$ and $r=8$, or 36,872 bytes in float32. The corresponding $d=256$ count is 4,610 parameters (18,440 bytes). These are head-only counts, excluding the encoder, replay, centroids, optimizer and runtime buffers. Centroid storage depends on the actual retained arrays; a 3,000-row fitting cap is not a centroid-count guarantee. Pure inference latency has not been measured for the complete protocol.

3.3 Training, metrics, and statistical boundary

The reported FT-Transformer configuration uses focal loss (gamma 2.0, alpha 0.75), learning rate 0.001, training batch size 384, evaluation batch size 512, and a negative-to-positive sampling ratio of four. At every checkpoint, evaluation covers all classes seen so far. Accuracy measures all correct predictions; Macro-F1 averages class F1 equally; balanced accuracy averages class recall; forgetting is the mean reduction from each earlier task's best historical accuracy to its later value. Current formal prediction values use seeds 1, 2, 3, 4, and 42. Model capacity and schedule are disclosed per dataset, and conclusions are descriptive unless a paired interval is explicitly reported.

3.4 Additive classifier-comparison protocol

TabM is added as a minimally invasive OFRA encoder comparison (Gorishniy et al., 2025). It produces 16 member embeddings of width 256 and averages them to the single representation required by the existing family-head interface. The family heads, exemplar budget, DP-Means routers, joint weight 0.5, cap 3,000, task order, and metric definitions remain unchanged. This is a mean-embedding adapter rather than a member-wise TabM-OFRA redesign.

CatBoost is evaluated separately as a cumulative multiclass diagnostic (Prokhorenkova et al., 2018). At each checkpoint it retrains on all training rows from classes seen so far. It does not use OFRA family heads, bounded replay, DP-Means routing, or ETG, and its native feature attribution is not the registered routed-margin SHAP target. The comparison therefore estimates classification headroom in the fixed feature representation rather than a matched continual-learning effect.

3.5 SHAP explanation protocol

The analyzer uses `shap.GradientExplainer`, an expected-gradients approximation, on the registered class margin $g_c(x)=\text{joint}_c(x)-\max_{\{f \text{ not equal } c\}} \text{joint}_f(x)$. The primary Malaya target is a CPU reconstruction evaluated on each fixed true-class probe batch, not an attribution of archived GPU score arrays. The wrapper has exact NumPy forward values with a detached correction and a differentiable surrogate backward path. Thus a same-batch forward pass and finite gradients do not prove gradient faithfulness. Positive/negative attributions refer to support for/opposition to the margin; top-k uses absolute magnitude. Fixed probes and backgrounds isolate checkpoint variation but do not test alternative splits or backgrounds.

For adjacent checkpoints, top-feature overlap is the intersection size divided by the union size of the two feature sets. The registered silent explanation-drift rule requires delta class recall > -0.05 and top-15 Jaccard < 0.70 . Thus the allowed recall loss is strictly smaller than five percentage points, not a symmetric stability band. Both thresholds are study-defined settings, not externally validated standards. The event unit is a class-by-adjacent-checkpoint transition, not a packet, flow or real-world incident.

The deletion diagnostic replaces selected features with frozen means and averages the decrease in class softmax probability. It is not the same numerical target as the attributed margin and can create unrealistic feature combinations. The random-control comparison is a limited faithfulness diagnostic, not a causal security test. Gradient-independent score perturbations and derivative checks remain necessary. Piecewise differentiability alone does not establish why an Integrated Gradients completeness check fails (Sundararajan et al., 2017).

3.6 ETG audit scope

ETG is the historical offline explanation-governance ledger. It records explanation-rule admission, refusal, drift escalation and one-step recertification. These action strings are simulated; no analyst response or security intervention is measured. Failed explanation admission does not cancel the underlying classification alert. The five-seed analysis consumes the CPU-reconstructed Malaya OFRA score described in Section 3.5, not a separately verified reproduction of archived GPU scores. It does not change future routing, training or replay.

3.7 Attribution-method robustness protocol

The source-bound robustness campaign uses Malaya checkpoints for seeds 1, 2, 3, 4, and 42, fixed true-class probes, the joint-cap-3,000 class margin, the top-15 rule, the registered random-control admission threshold, and the same ETG state machine. Its three primary rankings are Expected Gradients, single-feature ablation to the frozen checkpoint mean, and Gradient x Input. Pairwise top-15 Jaccard, admission agreement, ETG-state agreement, and transition-level silent-drift agreement are reported per seed and summarized with descriptive Student-t intervals. Integrated Gradients remain a failed single-seed numerical diagnostic and are excluded from the primary five-seed comparison. This design measures method dependence rather than assuming that any explainer is uniquely correct.

3.8 No-look-ahead audit protocol

The audit traces every data-dependent preprocessing state to its fitting split and task availability. It checks numerical accumulator reads and freeze points, categorical-vocabulary construction, official-test access, and row-level impact of future-only categories. The audit distinguishes future-category schema exposure from future-label training and from test leakage, then reports exact affected columns and rows rather than a binary pass/fail label.

3.9 Paired checkpoint-selection protocol

A ReplayIDS-aligned CIC-IDS-2017 D2 experiment compares last-epoch selection with two training-only checkpoint rules under the same fixed split, seeds {1, 2, 3, 4, 42}, FT256x4 encoder, replay-50 budget, training trajectories, and official joint-cap-3,000 inference. The unconstrained candidate selects the earliest maximum binary Macro-F1 on the training-only calibration holdout. The guarded candidate uses the same candidate checkpoints but admits a non-final checkpoint only when calibration Macro-F1 improves by at least 0.01, attack-recall loss is at most 0.01, and benign-FPR increase is at most 0.01; these are project-defined operating guards, not published universal thresholds. Heartbleed has one calibration row and therefore uses the registered last-epoch fallback. The five seeds are paired repetitions on one data split, not five independent datasets.

3.10 Same-data replay comparison and offline construction

Balanced Replay50 updates a shared FT-Transformer and multiclass head, pairing each later-task new row with one draw from old-class memory and retaining 50 herded examples per observed class. It is compared with same-seed OFRA joint-cap-3,000 results on Malaya and a separate ReplayIDS-aligned CIC-IDS-2017 D2 derivative. Splits, class supports, Task-0-only normalisation, and task order match within each dataset. Equal exemplar counts and epochs do not match total memory or compute: OFRA also stores centroids, replay changes row exposure, and the shared multiclass comparator differs from OFRA's binary focal loss and negative sampling. This is a system-level comparison, not an architecture-only ablation.

D2 caps normal-class training rows at the largest attack-class fitting count, 124,780. It contains 268,697 fitting rows, 68,313 training-only calibration rows, and 227,723 fixed test rows with 78 features. The cap uses counts from all fitting attack classes, including later increments. Thus D2 is an offline benchmark-construction policy, not a fully prospective no-look-ahead sampling policy. Official-test labels do not fit the cap, normaliser, or checkpoint selector. D2 results must not replace the distinct CIC-IDS-2017 results in Table 3.

Seeds 1, 2, 3, 4 and 42 vary training randomness on one fixed split and class order. Seed 42 participated in comparator selection; all-five estimates are descriptive, with seeds 1–4 reported separately. Last-epoch ReplayIDS OFRA is primary and guarded selection secondary. Post-hoc paired sign-flip tests use Holm correction across twelve listed metrics from Malaya and ReplayIDS, separately for $n=5$ and $n=4$; the smallest adjusted p is 0.75. For the specified exact two-sided sign-flip test, five nonzero pairs give a minimum p of 0.0625. This limit does not apply to every paired test, including a parametric t test. No confirmatory significance is asserted.

4. Results

Primary prediction findings are in Sections 4.3 and 4.9; Sections 4.1, 4.2 and 4.5 concern explanation and audit behaviour. Classifier headroom, open-set rejection, checkpoint selection and numerical portability are separately labelled diagnostics, not evidence of a single universally superior system.

4.1 RQ2: Explanation stability

Across five seeds, Expected Gradients reports 61 silent-drift events among 82 eligible class-by-adjacent-checkpoint transitions. The mean seed-level rate is 73.98% with a descriptive 95% Student-t interval of 58.38% to 89.57%. The earlier seed-1 value of 12/17 (70.59%) is retained only as a historical component of this aggregate. The evidence unit is a registered class transition, not a packet, flow, sample, or real-world incident.

The registered rule allows a class-recall decrease strictly smaller than 5 percentage points: $\Delta \text{recall} > -0.05$. This is one-sided, not a symmetric stability band. Drift is $\text{Jaccard}@15 < 0.70$. Feature Ablation reports 3/82 events and Gradient x Input 12/82. These method-dependent counts do not establish unsafe predictions, incorrect explainers or a fraction of drifting flows.

The five stored EG grids cover k in $\{5, 10, 15, 20\}$, Jaccard thresholds $\{0.5, 0.6, 0.7, 0.8\}$, and recall guards $\{0, 0.02, 0.05, 0.10\}$: 320 verified seed-setting rows. At $k=15$ and guard 0.05, mean seed drift rises from 3.69% to 48.20%, 73.98% and 96.40% across those thresholds. At the primary setting, the pooled ratio is 74.39%, distinct from the 73.98% mean seed rate. Across all 100 transitions, mean cosine similarity is 0.8525 and mean Kendall tau-b is 0.7686. Broad rank similarity can coexist with changes around a top-k boundary. These retrospective diagnostics do not calibrate an operational threshold.

4.2 RQ2: Attribution-method robustness

Across the five stochastic seeds, the three primary methods agree on admission for 50.0% of checkpoint-class rows on average, on ETG state for 54.0%, and on the silent-drift conclusion for 35.0% of adjacent transitions. Expected Gradients, Feature Ablation, and Gradient x Input have mean seed-level silent-drift rates of 73.98%, 3.60%, and 14.34%, respectively. Mean pairwise top-15 Jaccard is 0.516, 0.357, and 0.381 for the three method pairs. The result confirms material attribution dependence: ETG outcomes are method-conditioned, and the registered Expected-Gradients result is not treated as uniquely correct.

Table 2. Five-seed attribution-method agreement on the source-bound Malaya routed score.

Method pair	Top-15 Jaccard	Admission agreement	ETG-state agreement	Drift agreement
Expected vs ablation	0.516	68.7%	69.3%	40.0%
Expected vs Grad x Input	0.357	64.0%	69.3%	43.0%
Ablation vs Grad x Input	0.381	67.3%	63.3%	87.0%

Integrated Gradients are excluded from Table 2 because the earlier completeness diagnostic failed (mean row-level error 11.58; maximum 373.72). The three completed methods quantify sensitivity, not which attribution is causally correct. In particular, exact forward reconstruction does not resolve the surrogate-backward limitation of gradient-based results.

4.3 RQ1: Predictive performance and retention

Table 3. Current strict five-seed prediction evidence for the two main decision arms.

Dataset / model	Arm	Accuracy	Macro-F1	Balanced acc.	Forgetting
NSL-KDD FT256x4	Head only	56.29	31.08	36.48	8.14 pp
NSL-KDD FT256x4	Joint cap	71.56	41.46	42.25	1.60 pp
UNSW-NB15 FT256x4	Head only	68.78	21.16	23.52	2.68 pp
UNSW-NB15 FT256x4	Joint cap	61.95	23.50	28.45	8.97 pp
CIC-IDS-2017 FT256x4	Head only	64.59	21.25	25.91	9.04 pp
CIC-IDS-2017 FT256x4	Joint cap	72.57	36.95	63.13	9.63 pp
Malaya FT512x12	Head only	56.03	11.77	15.05	3.16 pp
Malaya FT512x12	Joint cap	54.68	21.15	23.03	3.55 pp
CSE-CIC-IDS2018 FT256x4	Head only	79.21	19.06	21.15	2.00 pp
CSE-CIC-IDS2018 FT256x4	Joint cap	50.16	33.55	52.15	19.07 pp

The routing effect is dataset-dependent. On NSL-KDD, joint cap 3,000 improves accuracy, Macro-F1, balanced accuracy, and forgetting. On UNSW-NB15, it improves Macro-F1 and balanced accuracy but lowers overall accuracy and increases forgetting. On CIC-IDS-2017, it improves accuracy and class-balanced metrics but does not reduce forgetting. On Malaya, the paired five-seed difference is -1.34 accuracy points, +9.37 Macro-F1 points, +7.97 balanced-accuracy points, and +0.39 forgetting points; pooled per-class results show non-zero recall for all ten classes under joint cap, but several minority recalls remain low. The table therefore supports dataset-specific trade-offs, not universal superiority.

The strict CSE-CIC-IDS2018 FT256x4 campaign completed five registered seeds with eight Task-0 epochs and ten epochs per later task. Head-only achieved 79.21% accuracy, 19.06% Macro-F1, 21.15% balanced accuracy, 2.00 points of forgetting, 31.70% attack recall, and 6.57% benign false-positive rate. Joint cap-3000 achieved 50.16%, 33.55%, 52.15%, 19.07 points, 91.82%, and 51.85%, respectively. Joint uncapped achieved 54.39%,

33.72%, 51.35%, 15.39 points, 84.38%, and 46.48%. Thus routing recovers minority/attack coverage but produces an operationally severe accuracy, retention, and false-positive cost on this dataset. The earlier 1+1-epoch closure remains historical and is not mixed with this result.

4.4 RQ1: Additive classifier comparison

Table 4. Malaya classifier comparisons. Values are mean +/- sample SD across five seeds; protocols remain separate.

Model and protocol	Accuracy	Macro-F1	Balanced acc.	Forgetting
TabM-OFRA joint cap, 5 seeds	57.99 +/- 3.93	20.46 +/- 1.19	21.60 +/- 1.70	4.39 +/- 2.56 pp
FT512-OFRA joint cap, 5 seeds	54.68 +/- 2.71	21.15 +/- 3.37	23.03 +/- 3.42	3.55 +/- 0.77 pp
CatBoost cumulative, 5 seeds	66.50 +/- 0.44	34.76 +/- 0.69	39.10 +/- 0.73	15.34 +/- 0.73 pp
TabM cumulative, 5 seeds	65.01 +/- 0.96	32.66 +/- 1.31	34.41 +/- 1.51	9.43 +/- 1.80 pp

Across the same five seeds, TabM-OFRA joint-cap accuracy is 57.99% versus 54.68% for FT512-OFRA, a +3.31-point difference. Macro-F1 is 20.46% versus 21.15% (-0.69 points), balanced accuracy is 21.60% versus 23.03% (-1.43 points), and forgetting is 4.39 versus 3.55 points (+0.84 points). The TabM change therefore improves aggregate accuracy but not the class-balanced or retention metrics. Because BitTorrent is the majority class, this result is not evidence of uniform minority-class improvement.

4.5 RQ3 ETG audit states and class performance

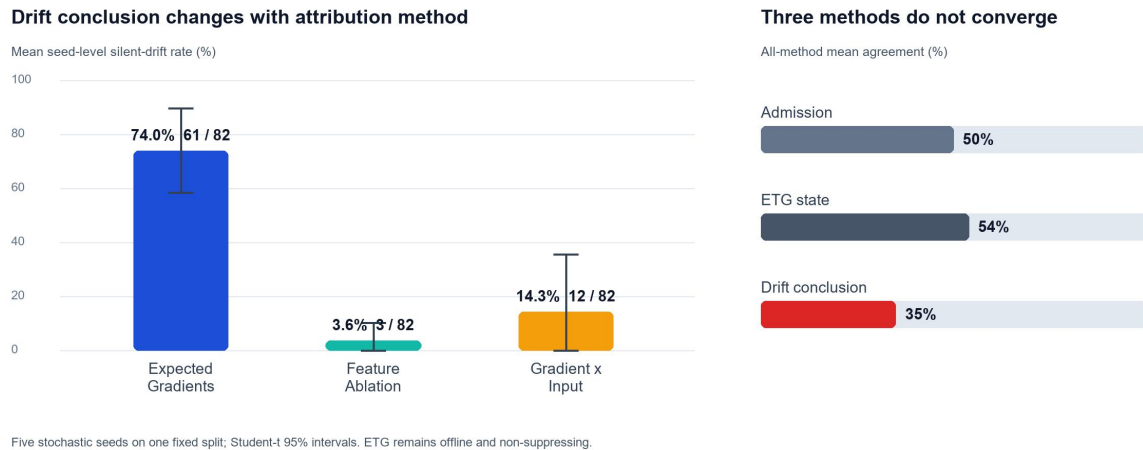


Figure 1. Five-seed attribution sensitivity and all-method agreement. Rates are method-conditioned audit evidence, not production incidents.

For Expected Gradients, the pooled five-seed ledger records 15 certified admissions, 35 refused admissions, 13 escalations, two strict recertifications, and six strict-recertification failures. Feature Ablation records 17 certifications and 33 refusals with no escalations, while Gradient x Input records six certifications, 44 refusals, one escalation, and one strict recertification. These differing state counts demonstrate execution of the same ledger under different explanations, but they also show that the governance outcome inherits the attribution choice. They do not quantify analyst benefit, false-alarm reduction, production remediation, or causal improvement of later OFRA checkpoints.

Retrospective enrichment binds all 450 method-specific rows to method identity, source-code and policy hashes, training artifacts and target scope. Protected protocols verify SHAP 0.51.0, NumPy 2.2.6, PyTorch 2.6.0+cu118 and probe-manifest hashes; custom methods are versioned by source hash. Across 150 common checkpoint-class records, state agreement is 42/72 below 20% recall, 19/43 at 20–50%, 5/15 at 50–80%, and 15/20 at 80–100%. Crucially, 27 of the 72 low-recall records have a historical CERTIFIED_STABLE state under at least one method. The label denotes an explanation-rule pass, not predictive reliability. Neutral display labels leave original states unchanged.

Replaying the hash-verified historical ledger reproduces all 450 primary states and actions. A further 240 retrospective replays vary Jaccard and recall guards at fixed top-15 across three methods and five seeds, retaining measured deletion admissions. Changed-k admission/state sensitivity requires new masked score evaluations. Neither a ledger action string nor a successful replay demonstrates human review, model repair or improved security outcomes.

4.6 Exploratory open-set and labelled-head pilot

The separate seed-42 O1 pilot used FTP-Patator as an unseen class before its declared labelled increment. Thresholds were fitted on 11,647 training-only known-class calibration rows; the test set was not used for

threshold selection. Confidence-only, centroid-distance-only, conservative joint, and empirical-joint rejection all produced 0% unknown recall. The best AUROC was 0.5303, the OSCR-style AUC was 0.4699, and the primary candidate buffer contained 509 known rows and no FTP-Patator rows. The registered gate therefore failed as an unknown-discovery mechanism.

After FTP-Patator labels were supplied at the legitimate task boundary, the ordinary supervised OFRA update achieved 82.30% new-class recall. Old-class accuracy changed from 87.28% to 87.01% (-0.27 percentage points). The evaluator never creates a semantic head automatically. This single-seed result supports labelled adaptation, not autonomous zero-day discovery, and remains an expected-contract reconstruction rather than an unchanged ReplayIDS reproduction.

Table 5. Seed-42 O1 open-set and post-label adaptation diagnostic.

Stage or metric	Result
Held-out class / test rows	FTP-Patator / 1,588
Pre-update known accuracy	87.28% (226,135 rows)
Unknown recall, all four gates	0.00%
Best AUROC / OSCR-style AUC	0.5303 / 0.4699
Post-label new-class recall	82.30%
Old-class accuracy change	-0.27 percentage points

4.7 Paired checkpoint-selection result

The training-only rule changes the operating trade-off but does not establish a stable improvement. Relative to last-epoch selection, final accuracy increases from 85.51% to 87.52% and Macro-F1 from 54.40% to 56.34%, while average task accuracy decreases from 79.78% to 77.88%, forgetting increases from 3.52% to 5.18%, and attack recall decreases from 85.73% to 83.76%. Benign FPR decreases from 13.10% to 10.44%. Every paired 95% confidence interval includes zero; the Holm-adjusted p-value for the two registered primary outcomes, Macro-F1 and forgetting, is 0.755. The candidate is therefore retained as an inconclusive ablation, and last-epoch selection remains the primary configuration.

Table 6. Per-class official joint-cap-3,000 results. Class-wise accuracy is recall; values are five-seed mean +/- sample SD on one fixed test split.

Class	Support	Last recall	Cal. recall	Last F1	Cal. F1
Benign	174,421	86.9% +/- 8.3%	89.6% +/- 5.0%	90.7% +/- 4.4%	92.1% +/- 2.8%
DoS GoldenEye	2,059	97.8% +/- 0.9%	94.7% +/- 4.8%	90.2% +/- 7.3%	92.5% +/- 2.0%
DoS Hulk	46,215	81.0% +/- 13.7%	81.1% +/- 18.1%	80.6% +/- 8.2%	82.7% +/- 7.5%
DoS Slowhttptest	1,100	75.0% +/- 14.3%	79.5% +/- 5.8%	54.6% +/- 7.7%	53.8% +/- 8.0%
DoS slowloris	1,159	69.0% +/- 21.2%	74.2% +/- 11.0%	48.3% +/- 16.5%	49.6% +/- 10.5%
FTP-Patator	1,588	69.0% +/- 44.3%	70.1% +/- 24.9%	30.9% +/- 28.5%	51.5% +/- 19.9%
Heartbleed	2	60.0% +/- 54.8%	60.0% +/- 54.8%	9.4% +/- 13.3%	7.6% +/- 10.4%
SSH-Patator	1,179	82.3% +/- 23.3%	69.0% +/- 38.7%	30.4% +/- 14.7%	21.0% +/- 16.6%

The per-class record explains why aggregate gains are not sufficient. Heartbleed has only two official test rows, so its mean and standard deviation are not a stable population estimate. FTP-Patator F1 improves under calibration, whereas SSH-Patator recall and F1 decline. The class-specific trade-offs reinforce the decision not to promote the candidate.

4.8 Guarded checkpoint-selection result

DICC Job 426307 repeated the five paired seeds with the recall/FPR guard. Relative to last epoch, guarded selection changed final accuracy from 85.51% to 87.10% (+1.59 points), Macro-F1 from 54.40% to 55.93% (+1.53), forgetting from 3.52% to 3.56% (+0.04), attack recall from 85.73% to 83.44% (-2.29), and benign FPR from 13.10% to 10.90% (-2.20). Average task accuracy decreased by 1.79 points. The Macro-F1 paired t-test gave $p=0.372$ and Holm-adjusted $p=0.744$; the forgetting test gave $p=0.977$. All paired 95% confidence intervals included zero. The guard removed the large mean forgetting penalty of the unconstrained selector but did not establish superiority, and only seed 2 Pareto-dominated its matched baseline across final accuracy, Macro-F1, forgetting, attack recall, and benign FPR. Last epoch therefore remains the formal primary rule.

Table 7. Guarded checkpoint selection versus the matched last-epoch baseline (five paired seeds).

Metric	Last epoch mean (SD)	Guarded mean (SD)	Paired change	95% CI
Average task accuracy	79.78 (6.86)	77.99 (10.01)	-1.79 pp	[-6.27, 2.69]
Forgetting	3.52 (2.60)	3.56 (4.71)	+0.04 pp	[-3.28, 3.36]
Final accuracy	85.51 (6.18)	87.10 (5.19)	+1.59 pp	[-0.33, 3.52]
Macro-F1	54.40 (5.07)	55.93 (5.02)	+1.53 pp	[-2.69, 5.75]
Attack recall	85.73 (14.98)	83.44 (17.49)	-2.29 pp	[-5.91, 1.33]
Benign FPR	13.10 (8.25)	10.90 (6.43)	-2.20 pp	[-4.87, 0.47]

4.9 RQ1 Same-data Balanced Replay50 comparison

Table 8 reports mean final metrics over five training seeds. Malaya OFRA has higher overall accuracy and lower signed forgetting in all five pairs. However, excluding screening seed 42, Macro-F1 is 20.70% versus 20.47% for replay, only +0.23 points; mean final task accuracy favours replay in all four new seeds. The large retention difference must therefore be read alongside weak absolute class-balanced performance.

Table 8. Same-data five-seed means. ReplayIDS uses last-epoch primary OFRA. Accuracy and recall values are percentages; forgetting is in percentage points.

Dataset / metric	Replay50	OFRA primary	Difference	Direction
Malaya accuracy	40.14	54.68	+14.54	Higher better
Malaya Macro-F1	18.72	21.15	+2.43	Higher better
Malaya balanced accuracy	22.13	23.03	+0.90	Higher better
Malaya signed forgetting	43.80	3.55	-40.25	Lower better
ReplayIDS accuracy	70.65	85.51	+14.86	Higher better
ReplayIDS Macro-F1	36.88	54.40	+17.52	Higher better
ReplayIDS balanced accuracy	87.22	77.63	-9.59	Higher better
ReplayIDS signed forgetting	5.89	3.52	-2.37	Lower better
ReplayIDS attack recall	99.78	85.73	-14.05	Higher better
ReplayIDS benign FPR	35.61	13.10	-22.51	Lower better

On ReplayIDS, primary OFRA improves overall accuracy and Macro-F1 and lowers benign FPR in all five pairs, but attack recall is lower in all five; forgetting improves in three. The test set is 76.59% benign, so fewer false alarms can increase overall accuracy while missing more attacks. Mean final task accuracy is 79.78% for OFRA versus 86.44% for replay. The four-seed sensitivity view retains the same central trade-off: accuracy 84.61% versus 71.71%, Macro-F1 54.07% versus 36.48%, attack recall 83.11% versus 99.77%, and FPR 13.76% versus 34.35%.

The same trained last-epoch OFRA states provide a mechanism diagnostic. Head-only Macro-F1 is 26.79%, router-only-cap-3,000 is 42.04%, and joint is 54.40%. Router-only attack recall is 92.06% versus joint 85.73%, while FPR falls from 21.76% to 13.10% when scores are combined. Thus routing contributes useful evidence, but score fusion changes the missed-attack/false-alarm balance. These retrospective scoring comparisons do not identify a unique causal component or justify optimising fusion weights on the test set.

4.10 ReplayIDS cross-environment score fidelity

A local CPU reconstruction checked all 20 stored primary checkpoint instances across five seeds against hash-bound historical GPU score snapshots. It used the original 512-row encoder chunks followed by whole-probe head/router scoring. The fixed gate required every score to satisfy absolute tolerance 0.0001 plus relative tolerance 0.00001 and required identical argmax predictions. Head scores passed at all checkpoints. The five initial checkpoints passed the combined gate, whereas all 15 post-initial Router/Joint checkpoints failed it. Maximum joint absolute error reached 0.01051009.

Two argmax differences occurred for the same probe row in seed 3 at checkpoints 001 and 003, changing DoS Hulk to Benign. They are two checkpoint evaluations of one row, not a full-test error rate or SHAP/ETG drift estimate. Reconstruction used CPU PyTorch 2.11.0 and NumPy 2.4.6; historical inference used A100 CUDA, PyTorch 2.6.0 and NumPy 2.2.6. The environments were not matched. Historical-environment GPU verification remains outstanding; no calibration was fitted, no historical score was replaced, and no revised full-test performance is claimed. This ReplayIDS result does not by itself re-evaluate the separate Malaya attribution campaign.

5. Discussion

5.1 When is the trade-off operationally worthwhile?

The value of routing depends on the dataset and operational cost model. NSL-KDD is favourable across the four displayed metrics. UNSW-NB15 trades lower accuracy and higher forgetting for better class-balanced metrics. CIC-IDS-2017 gains predictive coverage without a retention gain. Malaya improves coverage of minority application classes while its accuracy and forgetting differences remain uncertain. Such a trade may be worthwhile when a missed underrepresented class is more costly than a small aggregate-accuracy loss, but the decision must be based on per-class costs and cannot be inferred from overall accuracy alone.

The ReplayIDS replay comparison adds a distinct operational caution: OFRA reduces benign FPR but also misses more attacks in every paired seed. Whether that trade is acceptable depends on specified costs and minimum detection requirements. No such deployment threshold or analyst benefit has been validated here, so higher accuracy is not sufficient grounds to recommend deployment.

5.2 Classifier capacity and class imbalance

The encoder and cumulative comparisons answer different questions. TabM-OFRA tests a different encoder within the existing head/router interface. Cumulative CatBoost and TabM estimate headroom when all seen-class rows are retained and retrained. TabM-OFRA improves aggregate accuracy but not class-balanced metrics, while cumulative CatBoost improves final classification metrics and shows greater checkpoint forgetting. These observations do not identify model capacity as the sole cause. The same-data replay comparison further shows that isolation, loss, sampling, and memory accounting must be separated before attributing gains to an architectural innovation.

5.3 Meaning of the integrated evidence chain

The integrated evidence is an offline audit of a reconstructed score. Historical ETG state names and simulated escalation actions are retained for reproducibility, but governance is removed from the title and principal contribution. Until a matched-budget intervention study demonstrates benefit, ETG is an audit tool rather than a safety certificate or adaptive controller.

5.4 Attribution dependence

Three-method agreement is 50.0% for admission, 54.0% for ledger state and 35.0% for drift. Agreement alone cannot certify correctness, and passing a deletion rule can coexist with low class recall. The supplement records method-conditioned provenance, protected protocol dependencies and performance-stratified consensus. The next scientific requirement is independent faithfulness and utility evidence, not a more confident state name.

5.5 Unverified mechanism hypothesis

Old-class output changes may reflect changed parameters, class-set normalisation or competition from newly added rivals. A proposed four-score diagnostic separates these effects along a specified telescoping path. This is an order-dependent identity, not a unique causal attribution or a validated new algorithm. Old/new score-bias correction already appears in Scall and Weight Aligning (Belouadah and Popescu, 2020; Zhao et al., 2020), and GCR already identifies routing instability under expansion. The unresolved hypothesis is whether this decomposition helps allocate a fixed repair budget beyond random, periodic, error-only or disagreement-only triggers. No real-data repair result is presented.

6. Limitations and Threats to Validity

The five strict prediction campaigns use dataset-specific capacities and are not a matched-capacity comparison. The Malaya attribution campaign spans five training seeds on one split, with fixed background/probe choices, a CPU class-batch target and surrogate gradients. The recovered threshold and top-15 policy sweeps are retrospective and provide no external calibration. The three methods disagree materially; the prior IG diagnostic failed. ETG remains an offline audit without measured analyst or intervention benefit.

The current formal preprocessing uses Task-0-only numerical statistics and categorical vocabularies; the legacy 122/194-feature schema exposure was closed by strict cache rebuilding and affected five-seed reruns. Malaya is not an intrusion-detection dataset, and its application labels cannot support attack-recall or benign false-positive claims. Malaya is strongly imbalanced and several minority recalls remain low even after routing.

Cumulative CatBoost and TabM diagnostics are not matched continual-learning baselines. Complete runtime, explanation latency, energy, analyst burden, and protocol-matched external-baseline tests are not yet available across the full dataset suite. The FTP-Patator open-set result is a single-seed, single-held-out-class diagnostic; its 0% unknown recall invalidates the current gate but does not establish that all open-set methods will fail. The paired checkpoint-selection experiment uses five seeds on one fixed split; it does not establish cross-split generalization, and the Heartbleed calibration fallback prevents a fully uniform class rule.

Two further boundaries affect the newer comparison. ReplayIDS D2's training cap uses future-class fitting counts for offline construction, even though fitted normalisation is Task-0-only. The CPU reconstruction also fails the registered Router/Joint numerical-fidelity gate in a different runtime; source and checkpoint hashes alone therefore do not establish portable numerical equivalence. Same-environment verification is needed before further score calibration or attribution work on that ReplayIDS path. The balanced-replay results compare complete training systems with unmatched total memory and exposure and cannot isolate a human-designed component's causal contribution.

Within-dataset evaluation also does not establish cross-dataset transfer. CIC17-to-CIC18 testing requires a documented common feature/label mapping, compatible units and extraction semantics, source-only fitted transforms, and explicit treatment of target-only classes. Capture correlation, duplicates and collection artefacts can distort comparisons (El Mahdaouy et al., 2026). A common feature name alone is not evidence of equivalent measurements. No transfer result is claimed.

7. Conclusion

OFRA exhibits dataset-dependent prediction and retention trade-offs. The same-data replay comparison shows substantial Malaya retention differences but little Macro-F1 improvement in the four non-screening seeds; ReplayIDS accuracy gains coexist with lower attack recall. Existing threshold and three-method analyses show that explanation and ledger conclusions are strongly method-dependent. Low-recall classes can pass the historical explanation rule, so ETG is retained as an audit record, not an operational governance contribution. Matched baselines, numerical fidelity, attribution faithfulness and fixed-budget intervention tests remain necessary before claiming an algorithmic or deployment advantage.

Declarations

Computational resources: This research was supported in part through computational resources provided by the Data-Intensive Computing Centre, Universiti Malaya.

Ethics and data governance: The study uses previously collected public benchmark or research datasets and does not involve direct interaction with human participants. Dataset licences and institutional computing rules remain applicable. No restricted credentials or private institutional data are included in the public repository.

Code and data availability: Reproducibility code, configurations, preprocessing contracts, result summaries, and public evidence bindings are available at <https://github.com/1838904818/Class-IL>. Original datasets remain with their respective providers; the repository documents acquisition and preparation rather than redistributing data where licence or size makes redistribution inappropriate.

AI assistance: AI-based tools assisted code development, technical analysis, and manuscript drafting and revision. These tools are not authors. Responsibility for the research decisions, verification of results, interpretation, and final manuscript remains with the author. The contribution claims concern the specified design and empirical evidence, not an assertion that all constituent methods or ideas were created without assistance.

References

- Adabara, I., Sadiq, B. O., Aliyu, N. S., Yale, I. D., Maninti, V., and Mutebi, J. (2025). Toward trustworthy cyber defense: A cognitive intrusion detection system with explainability and probabilistic uncertainty calibration. *KIU Journal of Science, Engineering and Technology*, 4(2), 160-172. <https://doi.org/10.59568/KJSET-2025-4-2-16>
- Belouadah, E., and Popescu, A. (2020). ScalL: Classifier weights scaling for class incremental learning. *WACV*. <https://arxiv.org/abs/2001.05755>
- Bhat, P. S., Yazdani, S., Arani, E., and Zonooz, B. (2026). Parameter efficient continual learning with dynamic low-rank adaptation. *TMLR*. <https://arxiv.org/abs/2505.11998>
- Chae, J., et al. (2026). GCR: Geometry-consistent routing for task-agnostic continual anomaly detection. *arXiv preprint*. <https://arxiv.org/abs/2601.01856>
- Cossu, A., Spinnato, F., Guidotti, R., and Bacciu, D. (2024). Drifting explanations in continual learning. *Neurocomputing*, 597, 127960. <https://doi.org/10.1016/j.neucom.2024.127960>
- El Mahdaouy, A., Ait Yahia, I., Oualil, S., and Berrada, I. (2026). Deep learning for contextualized NetFlow-based network intrusion detection: Methods, data, evaluation and deployment. *arXiv preprint*. <https://arxiv.org/abs/2602.05594>
- French, R. M. (1999). Catastrophic forgetting in connectionist networks. *Trends in Cognitive Sciences*, 3(4), 128-135. [https://doi.org/10.1016/S1364-6613\(99\)01294-2](https://doi.org/10.1016/S1364-6613(99)01294-2)

- Garcia-Santaclara, P., Fernandez-Castro, B., and Diaz-Redondo, R. P. (2024). Overcoming catastrophic forgetting in tabular data classification: A pseudorehearsal-based approach. *arXiv preprint*. <https://arxiv.org/abs/2407.09039>
- Goncalves, H., and Alomari, Z. (2026). Explainable AI for cloud intrusion detection: A user study of SHAP and LIME in AWS GuardDuty. 2026 IEEE 5th International Conference on AI in Cybersecurity. <https://doi.org/10.1109/ICAIC67076.2026.11395766>
- Gorishniy, Y., Kotelnikov, A., and Babenko, A. (2025). TabM: Advancing tabular deep learning with parameter-efficient ensembling. International Conference on Learning Representations. <https://arxiv.org/abs/2410.24210>
- Gorishniy, Y., Rubachev, I., Khruikov, V., and Babenko, A. (2021). Revisiting deep learning models for tabular data. *Advances in Neural Information Processing Systems*, 34, 18932-18943. <https://proceedings.neurips.cc/paper/2021/hash/9d86d83f925f2149e9edb0ac3b49229c-Abstract.html>
- Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., and Chen, W. (2022). LoRA: Low-rank adaptation of large language models. International Conference on Learning Representations. <https://openreview.net/forum?id=nZeVKeeFYf9>
- Kirkpatrick, J., et al. (2017). Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences*, 114(13), 3521-3526. <https://doi.org/10.1073/pnas.1611835114>
- Kulis, B., and Jordan, M. I. (2012). Revisiting k-means: New algorithms via Bayesian nonparametrics. *Proceedings of the 29th International Conference on Machine Learning*. <https://icml.cc/2012/papers/291.pdf>
- Lamaakal, I., Yahyati, C., El Makkaoui, K., Ouahbi, I., and Maleh, Y. (2026). Drift-to-action controllers: Budgeted interventions with online risk certificates. CAO Workshop at ICLR. <https://arxiv.org/abs/2603.08578>
- Lundberg, S. M., and Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 30.
- Maseno, E. M., Sun, Y., and Wang, Z. (2026). Detecting adversarial evasion in deep learning intrusion detection systems using explainable AI. *International Journal of Information Security*, 25, Article 125. <https://doi.org/10.1007/s10207-026-01299-x>
- Nugraha, B., and Bauschert, T. (2026). E2D2: A modular framework for explanation-based drift detection in unsupervised network intrusion detection. 2026 IFIP Networking Conference. <https://doi.org/10.23919/IFIPNetworking70592.2026.11578962>
- Prokhorenkova, L., Gusev, G., Vorobev, A., Dorogush, A. V., and Gulin, A. (2018). CatBoost: Unbiased boosting with categorical features. *Advances in Neural Information Processing Systems*, 31.
- Qiu, Z., Zhou, D., Zhai, Y., Liu, B., He, L., and Cao, J. (2024). VAEMax: Open-set intrusion detection based on OpenMax and variational autoencoder. *arXiv preprint*. <https://arxiv.org/abs/2403.04193>
- Rebuffi, S.-A., Kolesnikov, A., Sperl, G., and Lampert, C. H. (2017). iCaRL: Incremental classifier and representation learning. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2001-2010.
- Sundararajan, M., Taly, A., and Yan, Q. (2017). Axiomatic attribution for deep networks. *Proceedings of ICML, PMLR 70*, 3319-3328. <https://proceedings.mlr.press/v70/sundararajan17a.html>
- Xie, X., Qiu, Y., Lin, R., Zheng, W., and Wang, R. (2024). Class incremental learning with task-specific batch normalization and out-of-distribution detection. *arXiv preprint*. <https://arxiv.org/abs/2411.00430>
- Zhang, X., Bai, L., Yang, X., and Liang, J. (2025). C-LoRA: Continual low-rank adaptation for pre-trained models. *arXiv preprint*. <https://arxiv.org/abs/2502.17920>
- Zhao, B., et al. (2020). Maintaining discrimination and fairness in class incremental learning. *CVPR*. <https://arxiv.org/abs/1911.07053>